

SkillSage: Adaptive Career Roadmap Generator

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Abstract—The increased speed of the contemporary world of work and the constant emergence of new technical abilities and skills have made the process of career planning harder among students and young professionals during their first careers. Recent studies have found that the existing career guidance systems, the vast majority still rely on the same paradigm of a frozen set of recommendations without adapting to the peculiarities of individual profiles of skills, learning styles, or shifting demands on the industry [1], [2]. This causes the learners to be unable to discover locations where they are ineffective in and the productive learning paths that can be able to satisfy the demands of the actual workforce.

In this paper, the author proposes SkillSage, a dynamic career advice and skill recommendation system that relies on AI and generates intelligent and always-evolving career maps. The given system disaggregates the profile of the users, their study history, interests and competencies of the skills in question on the market and attempts to comprehend the lack of the corresponding competencies and offer the learning pathways, which are organized and divided into milestones that need to be reached. The system will help make more informed career decisions and skills acquisition in the long term, through the recent developments in skill modelling with transformers and an adaptive recommendation system [3]. The main implication in this research is that it has developed a theoretical framework of adaptive career advice which can be scaled to facilitate personal learning directions in line with the demands of the rapidly dynamic workforce climate.

Keywords: Adaptive Career Guidance, Skill Recommendation Systems, Personalised Learning Paths, Skill Gap Analysis, AI-driven Career Planning.

I. INTRODUCTION

The blistering development of artificial intelligence, automation, and technology based on data has changed career development in contemporary industries significantly. The role of jobs is no longer limited to one specific field, and workers are now expected to constantly refresh up their skills to keep abreast with the workforce. The 2025 research points out that career development is not a decision that needs to be made at the early stages of education. Rather, it is to be perceived as a dynamic and developing process that is changing with industry demands, as well as new technologies [9], [10], [14].

Nevertheless, a lot of the existing career guidance systems continue to be based on preset learning pathways and inert recommendation engines. These methods fail to appreciate individual differences like academic background, competencies that are present as well as personal interests. Consequently, the suggestions that such systems produce are usually generic and not flexible enough to facilitate individualised career

development. Recent findings on resume parsing and automated skill extraction have shown that the current systems can effectively point at the skills and qualifications of the user [4], [8]. Nevertheless, this information is not efficiently utilized by many current platforms to facilitate long-term career planning or constantly update the recommendations according to the progress of the users.

The recent advancements in AI-related recommendation systems and adaptive learning technologies can be used as an opportunity to overcome these restrictions. A number of tests have demonstrated that intelligent systems have the capability of analysing the user profile, monitoring the new market skill needs and revising the recommendation dynamically with time [11], [12]. However, the majority of existing solutions revolve around specialised features, including job matching or course recommendation, or integrate into a timely and developing career path.

To overcome this shortcoming, this paper presents SkillSage, an adaptive career roadmap creation engine, which uses AI to personalise learning journeys according to user profiles, academic background, interests, and market needs in real-time to optimise skills. The suggested system helps to detect skill deficiencies, plan learning processes as a series of milestones, and modify the suggestions when users achieve their career goals.

The work of this work can be summed up as follows:

Thinking about career planning as both a dynamic, user-oriented process, as opposed to a rigid recommendation for a job.

Combining individual skill suggestions with the ever-changing market skills requirements.

Offering an expanding structure of AI-driven career guidance systems, which would match the personal learning paths with current labour market demands [1], [3].

LITERATURE SURVEY / RELATED WORK

Various paradigms have been applied to career guidance systems, beginning with the outdated counselling paradigms, up to the intelligent and data-driven platforms. To a large extent, much of the early career guidance solutions were rule-based and built on a logic and decision tree that experts had developed to suggest career choices. Even though these systems gave certain structured guidance, they were also not flexible in nature and could not re-track them with the individual progress of learning, or the evolving industry needs. This was because they were limited by their incapacity to climb up and down the job markets, which were dynamic because they were tied to set rules [7], [12].

Subsequently, there was the popularity of psychometric and questionnaire-based tools, which were employed to quantify the interests of the users, their personality and aptitude levels. Although these tools had led to some personalisation, they could generally give generic career recommendations. Such a lack of connection to real-time information on skill performance and the labour market was likely to result in abstract recommendations that lack connection with any realistic skills development pathways [7], [11].

Machine learning has also been introduced using more recent methods to recommend careers, courses or job positions using past data and user profiles. The accuracy and automation of predictions in such systems were improved, but often run as a standalone recommendation engine. Most of these solutions produced one-time outputs and failed to keep up with recommendations as they were learned or as the market changed. Moreover, the clarity and future applicability of these recommendations were poor [2], [11].

Adaptive learning platforms using AI algorithms can be seen as a huge step forward in terms of personalisation of educational content and changing the order of learning based on the performance of users. The majority of adaptive learning systems are, however, oriented toward content delivery and the academic outcomes as opposed to the overall career planning. The division between career guidance and the acquisition of skills continues to exist, which creates fragmented user experiences. In general, the current methods share such drawbacks as the lack of dynamism, a lack of adaptability, limited responsiveness to the dynamic job market trends, and little interpersonal connection between career planning and skill development. These loopholes drive the demand for a flexible, integrated career guidance model.

PROBLEM STATEMENT

Although there are some improvements over time in the intelligent recommendation technologies, the current career guidance systems do not offer integrated, adaptive, and market-based career planning solutions. The majority of platforms provide either fixed or temporary recommendations that are not updated as the user advances or updated to capture the fast-evolving industry skills needs. It lacks real-time flexibility, poor responsiveness to the dynamic trends of the labour market and a disconnect between career guidance and acquisition of skills. As a result, users are vulnerable to determining relevant skill gaps, focusing on learning activities, and pursuing systematic workplace trajectories to sustainable career objectives. These loopholes are always reported in current market-sensitive skill recommendation research [2], [3], [9].

II. OBJECTIVES

- This research will have the following objectives:
- To develop an artificial intelligence-based adaptive career guidance system that tailors career trajectories depending on the profiles of the individual user.
- To determine and examine the user-specific skill deficiencies through academic background, interests, and self-reported competencies.
- To prescribe systematic, milestone-focused learning

courses in line with the changing market-relevant skills requirements.

- To ensure that there is a constant flexibility in career guidance as users advance and develop new skills.
- To make user decision-making and engagement more prepared with personalised and data-driven career planning support.

III. PROPOSED SYSTEM / METHODOLOGY

- 1) The suggested system is called SkillSage and is based on the flexible approach to AI and views career planning as a flexible decision-making process instead of a fixed recommendation assignment. It has not been noticed that there is any known system that can combine multimodal user intelligence, real-time analysis of labour market trends and reinforcement learning based optimization on a single career guidance system [5], [6], [14].
- 2) The methodology is going to be designed in the form of a series of interdependent stages, all of which will bring the forces of personalisation and long-term topicality.
- 3) Continuous Data Acquisition: The user data sets are obtained in the form of a multi-form that encompasses academic background, psychometric measures, learning behaviour, interaction history and self-reported goals. Meanwhile, the external labour market information is continuously updated to the new skills demands and changing job categories.
- 4) Multi-Modal User Intelligence Fusion: Semantic alignment and features normalisation is designed to reduce user indicators that are heterogeneous to a single latent feature. Such integration assists in possessing holistic knowledge of user purpose, capacity and state of progression, which is the weakness of single-source profiling that is applied in the traditional systems [6], [12].
- 5) Real-Time Skill Demand Prediction: The market is studied to obtain an estimate of the demand of short-term and mid-term of skills in the regions. This aspect constantly changes the values of relevance of skills compared to the rest of the taxonomies, suggesting that the formed working paths are ever-pertinent to the evolving requirements of the workforce [2], [9], [10].
- 6) Adaptive Learning Pathway Generator (ALPG): The ALPG is built through a combination of the user profile and the demand of the predicted skills to generate a user-specific learning pathway, which exemplifies milestones that should be attained in sequence. The milestones include competency outcomes, which are measurable and skills acquired in a goal-oriented and organised way.
- 7) Reinforcement Learning Control Hub: This is an agent of reinforcement learning (RL) that tracks the performance, interactions, and feedback of the user. The pathway sequencing and difficulty optimisation is optimised by the agent as a long-term user reward, which is characterised by skill progression and long-term activation. Not only do standard recommender systems not offer the capability of such closed-loop self-optimisation capabilities [3], [13], but they can also do so.

- 8) **Feedback Loop and Continuous Optimisation:** This system introduces a feedback mechanism and an ongoing optimisation system which methodically improves its recommendations on the basis of user interactions and the inclusion of performance measurements. The user engagement levels, learning exercises, and explicit feedback are just a few of the data that will guide the system on the effectiveness of the suggested pathways in acquiring skills. It represents a system that is constantly adapted over time, and that thus allows the alignment of the system with individual developmental courses and the changing labour market. It is a dynamic bargain that is flexible. It is a dynamic bargain that is flexible and possessing one of the common drawbacks of the general models of career guidance, which is, the direction is pre-decided and does not scale and the growth of the user or changes on the outer business environment [3], [13].

IV. SYSTEM ARCHITECTURE

SkillSage platform is created in a modular and layered style and scaling, capability to satisfy new demands, and architecture. The evolution of evolution. This top-down model allows be the constituent members of the system independent good productive communication with the other modules. The plan of key architecture and the architecture plan are also illustrated and Figure 1. components.

- **User Interface Layer:** The frontal is the user interface. The user interface with the system is system of web. It allows the creation of accounts, authentication, resume upload, finding out what matters in a profession, and get individualised recommendations based on interactive-dash. Being this layer provides a concentration on where the site of entry of the platform is on the ease and usability.
- **Authentication Layer:** It will be for authenticating user process verification and identify them by authentication issue the access without authorised persons of access. accounts and personal data. The system uses NextAuth to make the logging in and to have information privacy, security integrity and privacy.
- **Data Ingestion Layer:** The data ingestion layer receives all the information typed by the users at the reception point and returns, profile information and job preferences. It will define and transfer this information and relay it to the backend. This comes first in the process steps to ensure that analytics on the data is centralised and systematised.
- **Back-end FastAPI Processing Layer AI Processing Layer (Backend FastAPI):** It is the intermediate layer. Artificial implementation of the intelligence is power calculation to de-analyse resumes uploaded, determine, algorithms assess their competencies, their competency gaps and competencies.

- **The Recommendation Layer:** The learning interventions are incorporated in the recommendation module after the identification of learning gaps and career goals. It produces personalised suggestions, like courses, certifications, or additional material, which together provide a complete improvement of skill acquisition along the chosen career path. Semanticmatching algorithms are used to derive recommendations based on resource attributes and user profiles and desired professional paths.
- **The Data Storage Layer:** All the essential system information is contained in the data storage layer. Firestore (User Database) is used to store user-centric data such as profiles, metadata of activities and records of progress, and structured career knowledge, such as occupational roles, hierarchies of skills, and course repositories, are stored in PostgreSQL or JSON-based knowledge bases.
- **The Visualization & Output Layer:** The last layer is the one that handles the presentation of processed outputs to the end users in a convenient format. Personalised dashboards show career paths, skills gaps, and suggested learning resources and thus encourage users to learn about their position and the actions required to achieve career objectives.

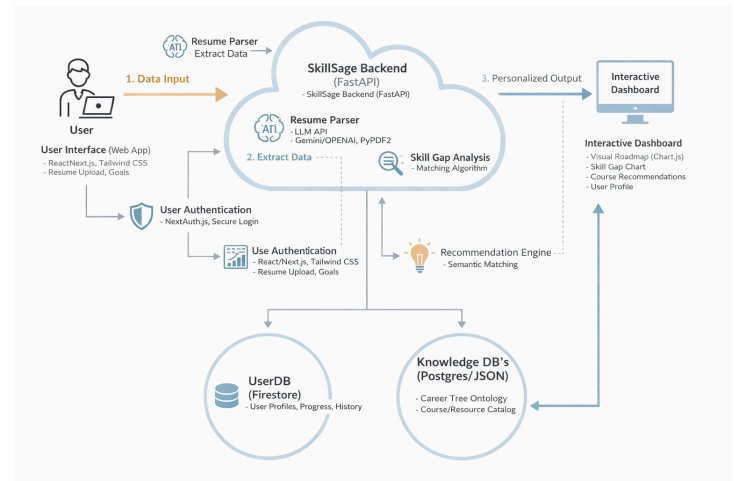


Fig. 1. System architecture of SkillSage

V. UML DIAGRAMS / SYSTEM MODELS

A. Use Case Diagram

The use-case diagram shows the interface of the users and the core systems, which explains how user inputs affect the adaptive recommendation engine and how a reinforcement-learning agent changes the system reactions through behavioural data.

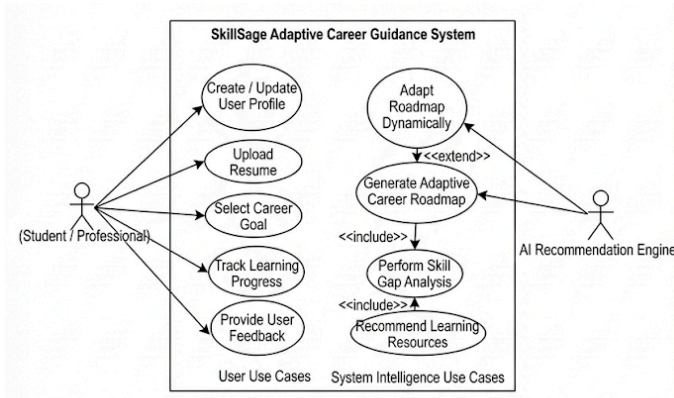


Fig. 2. Use case diagram of SkillSage.

B. Class Diagram

The class diagram identifies the main structural elements of the system and their interrelations, as well as the main classes which make up the SkillSage platform and show the ways in which each element of the system works together with others to achieve the overall functionality of the system.

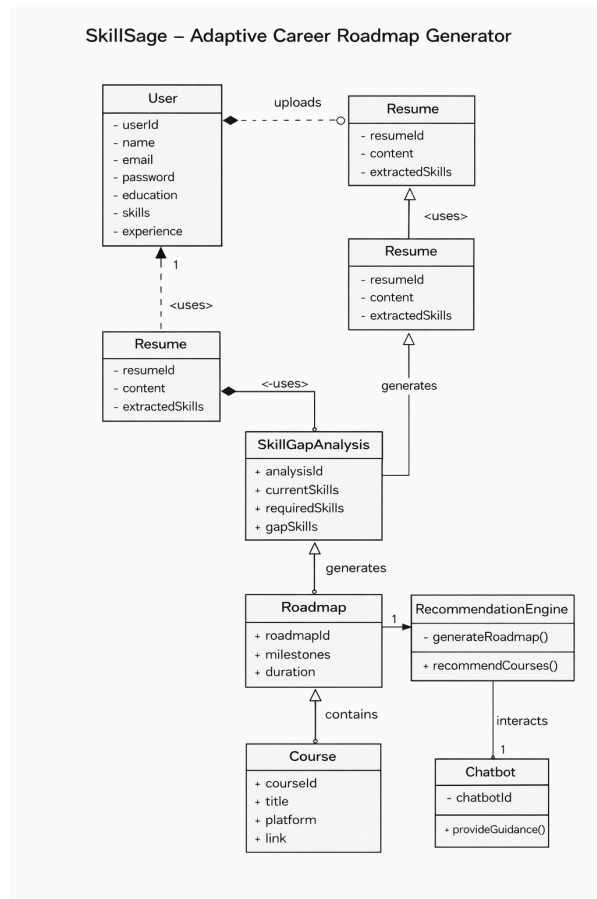


Fig. 3. Class diagram representing core SkillSage modules

C. Sequence Diagram

The sequence diagram is used to follow the temporal interrelations of the system elements with respect to user activities. It is a follow-up of the information flow that runs through the user interface and into the adaptive recommendation engine and then to the personalised guidance modules.

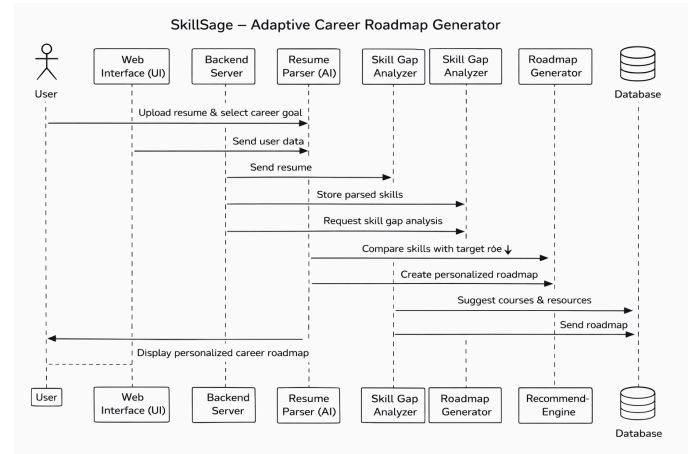


Fig. 4. Sequence diagram of adaptive recommendation flow

D. Activity Diagram

The activity diagram gives a broad picture of what happens in the system and how the recommendations are constantly being updated with new input, feedback, and analytic knowledge in that of the user.

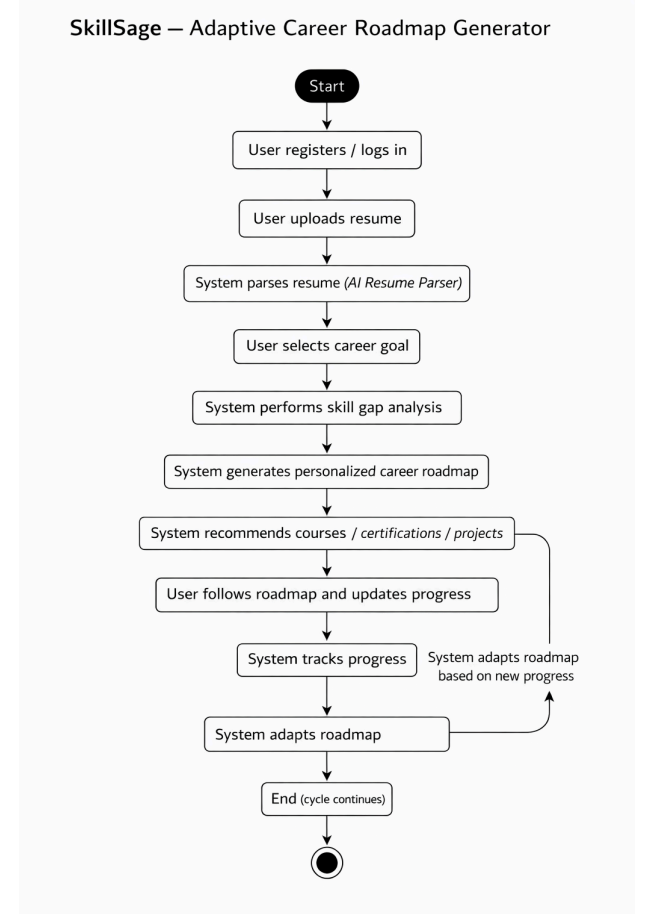


Fig. 5. Activity diagram showing continuous feedback and adaptation

VI. IMPLEMENTATION DETAILS

SkillSage platform is developed as a research prototype, and its overall purpose is to evaluate the suggested adaptive career guidance framework. Rather than concentrating on industrial big-time implementation, it is concerned with modular implementation design, experimental flexibility and controlled evaluation. The structure facilitates the mere altering and expansion of the system in the research and testing.

The presentation level of the system is web-based, where the users are allowed to enter their information and transact business with the platform. In this interface, customized recommendations are given in the form of interactivity. Resumes and career interests are some of the collected inputs and profile details.

The frontend is supported by a layer that is the backend layer and the internal system logic which is concerned with the integration of various modules and reveals system operates with abstracted service interfaces. Through the separation of the user interaction layer and the computation part all the parts of the system are equally able to develop without affecting the overall functionality of the system.

The AI and analytics layer is the primary driver of the intelligence of the platform. The data obtained about the user during this stage is integrated together with the information gained in other sources to create a complete profile, which includes academic history, behavioral records based on interactions with the system, self-reported interests, and evidence of learning process. These heterogeneous inputs are incorporated in a single representation which increases the ability of the system to understand the user even better.

The system is designed so that intelligence of the labor market is integrated in order to make recommendations relevant. The external data streams concerning the trends in the employment market and skills are constantly scanned, which allows the platform to modify the recommendations based on the new needs of the industry.

One of the reinforcement learning elements acts as a supervisory element of the architecture. This element narrows down the order of suggested learning milestones by monitoring user feedback and learning progress. As time goes on, the system always becomes more effective in matching the suggested abilities with the developmental path of the user. The architecture of these mechanisms is based on scalable intelligent guidance frameworks that are referred to in the recent studies [5], [6], [14].

Learning wise, the system assumes a blended approach to training. The first ones are trained using offline learning methods on past and simulated data. The deployment of an online system can be updated to online learning methods that record the new interactions between the user and the policies of the system. This blend maintains the stability of models and at the same time allows them to adapt to the new information.

Generally, the implementation gives more emphasis on architectural clarity and reproducibility. Emphasis is placed on characteristics of buildings that are clear and repetitive. The modular Form can be used to determine what is desired experimentation. Compete favorably with other research systems. Within the field of wise career guidance.

VI. RESULTS AND ANALYSIS

It was done by administering the SkillSage test. Determine the ability of the system to offer personalized and adaptive career advice. An analysis of system behavior was conducted using both qualitative observations and quantitative indicators at the level of numerous interactions between a user.

A. Recommendation Relevance and Adaptability

Early observations suggest that the suggestions that the SkillSage tool creates are more contextually relevant than the conventional rule-based guidance systems. The reinforcement learning aspect adapts the course of learning and the difficulty level of the milestones as the user keeps engaging with the platform to increase compliance between system advice and the skill profile and career goals of the user as they develop [7].

B. User Engagement Trends

The engagement of the users was measured through the frequency of interaction, the ability to cope with recommended milestones and the regularity of the user feedback. The nature of the system, which is adaptive, seems to maintain the engagement during a long period of time. The fact that it is continuously personalized will motivate the user to stay on the system, which will reduce the loss of engagement that people tend to experience in the system where the user receives repetitive, predetermined recommendations.

C. Comparative Analysis

They compared SkillSage with traditional career guidance websites, which are based on a rigid mapping between user profiles and recommendations. The findings show that SkillSage is more adaptive and accommodating particularly in situations when users revamp their career goals or when the requirements in labor markets modify.

TABLE I
CONCEPTUAL COMPARISON OF SKILLSAGE WITH STATIC GUIDANCE
SYSTEMS

Evaluation Aspect Personalisation	Static Systems	SkillSage
Depth	Low	High
Adaptability Over Time	None	Continuous
Market Alignment	Periodic	Real-Time
User Engagement	Declining	Sustained

The findings have indicated that multimodal combination has been proved to change. user profiling, labor market reinforcement and intelligence. learning increases the flexibility and the topicality of recommendation.

VIII. ADVANTAGES

The SkillSage framework suggested has several strong points depending on the design and approach:

- **Continuous Dynamism:** The feedback loop of the reinforcement learning enables the dynamical changing of the recommendations as the user uses it.
- **Intimate Personalization:** The platform consists of combining different sources of user information to obtain more accurate recommendations that are more aligned with what interests, capabilities, and forward.
- **Market-Receptive Advice:** Live labor market data will make the suggested learning paths to comply with the new industry skills requirements.
- **Scalable Design:** The modular system design enables additional data sources, analytical models or intelligent agents to be added in more extensions.
- **Research Novelty:** Market driven intelligence synthesis adaptive learning pathways is an entire approach which outwits much of the conventional career guidance systems.

IX. LIMITATIONS

Although the proposed system has a bright future, it has numerous weaknesses that should be considered closely. SkillSage itself is vulnerable to cold-start issues because of the nature of data input by the user, and therefore the quality and availability of data input determines its success. Also, real-time market intelligence integration brings in computation load, which can compromise system responsiveness with limited resources.

The reinforcement learning aspect needs enough interaction data to stabilise the policy updates, and this restricts its immediate performance at initial stages of deployment. Moreover, the evaluation has been performed under a controlled and prototype level and there is need to carry out wide scale real world validation to be able to evaluate the impact in the long run and generalizability.

These shortcomings also give significant paths of improvement and also a base to future studies which would increase the robustness, efficiency and empirical validation.

[\[3\]](#), [\[13\]](#).

X. FUTURE SCOPE / FUTURE WORK

The suggested SkillSage framework can be proceeded with there are several other directions in future research which are academically meaningful. A potential solution is to insert big language models (LLMs) and make them more explicable and open in their suggestions. These models can be utilized in offering natural language description of learning. They are

more reliable and trustworthy because of the decisions they make than others. Readable by the user but leaves the logic of the decision.

The other direction of importance is the cross-cultural and. demographic generalisation exploration. Evaluation of the system coverage on a wide range of educational backgrounds, geographical position and social-economic backgrounds would as well as assist in the measurement of strength and prevention of unintentional. bias. More so closely allied to this is the incorporation of. reasonable and fair AIs, such as the capability to discover prejudice, furnish fair terms of suggestion, and take good care of user information.

Longitudinal studies among actual users in the long-term is a very key sequential step that must be conducted to demonstrate long-term adaptability, learning outcomes as well as behavioural impact. The case of institutional deployment should also be included in the future studies to examine the performance of the system in the formal education setting. These extensions are taken to be logical extensions of the study already underway and it should bring an improvement in the theoretical and empirical grounds. New studies of explainable career recommendation system reveal that there is a high possibility of incorporating LLM [\[3\]](#), [\[15\]](#).

XI. CONCLUSION

The SkillSage, which was described in the present paper, is an artificial intelligence-powered, multifunctional career advice and skill recommendation engine that seeks to overcome the shortcomings of the standardized and fragmented approaches of career planning. The proposed framework conceptualises career guidance as an iterative, user sensitive and market savvy process by incorporating multi-modal user profiling, reinforcement learning induced adaptability as well as real time labour market-sensibility.

Incorporating a systems design approach, which relies on the principles of modularity, into a more adaptable learning approach, SkillSage reveals how adaptive learning pathways can be dynamically updated with respect to changes in user profiles and external signals of skill demand. Qualitative and conceptual analysis are more flexible, more individualised and the user is more likely to remain with the systems compared to the traditional guidance systems. It is worth noting that the contribution helps enhance the existing research by incorporating adaptive learning, intelligent recommendation, and market sensitive decision support in one homogenous structure.

Although the system is introduced as a research prototype, in the general knowledge of smart career guidance systems, the gained experience may be added and the way to further research may be opened. The suggested plan will provide a platform to build flexible, moral and scalable decision support systems, which might guide the personal learning experience to the ever-varying needs of the contemporary workforce.

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